No.	Solution
1(a)(i)	Acceleration of the ball $a = g\sin\theta$ $= 9.81\sin 40^\circ$ $= 6.306$ $\approx 6.31 \text{ m s}^{-2}$
1(a)(ii)	For uniformly accelerated motion, use $v^2 = u^2 + 2as$ $v^2 = 2(6.306)(0.56)$ $v = 2.658$ $\approx 2.66 \text{ m s}^{-1}$
1(b)(i)	Centripetal force acting on the ball $F_c = \dfrac{mv^2}{r}$ $= \dfrac{(72\times10^{-3})(1.5)^2}{0.12}$ $= 1.35 \text{ N}$
1(b)(ii)	For the ball at the top of the loop $N + mg = F_c$ $N = F_c - mg$ $= 1.35 - (72\times10^{-3})(9.81)$ $= 0.6437$ $\approx 0.644 \text{ N}$ Magnitude of the force due to the track acting on the ball = 0.644 N Direction of the force is vertically downwards towards the centre of the circular loop.
2(a)	Gravitational potential at infinity is taken to be zero. Due to the attractive nature